Join our community, grow your knowledge and learn from others!

Sign In

Don't have an Epic Games account? Sign Up

Lobbies Interface

Create and manage online lobbies.

Users to use this **Beta** feature, but use caution when shipping with it.

The **Online Services Lobbies Interface** provides an API for creating and managing lobbies. A **lobby** is a group of players within the context of a client application where all of the players have some shared state. This could be used to allow all of the players in the same online match together or all waiting for the same match to begin. Both the lobby and its members have attributes used to share this state. The Lobbies Interface provides a set of functions for the addition and modification of lobbies and lobby members in real-time.

The lobby leader may change a lobby's attributes, and each member (including the leader) may change their own member attributes. The lobby owner may also change lobby member attributes, along with their types and value constraints, that are then used to create a lobby. This lobby schema is defined in the configuration and lobby files and is validated by the lobby service. Lobbies are required to define all attributes to create a lobby, but if any attribute or member attribute fails schema validation, lobby creation will fail.

Lobbies can optionally be a lobby group, which supports together lobbies proceeding into a match. Based game matches may be played within the lifetime of a single lobby. Below is an example lifecycle of a lobby:

- A player creates a lobby with the desired privacy settings and attributes. This player is designated the lobby leader.
- Additional players locate and join the lobby based on search, invitation, or social presence.
- Players share attributes within the lobby through attribute updates.
- Once the lobby chooses to play a match, the lobby leader records the match session ID as a lobby attribute.
- Other lobby members see this session ID and join the same game session.
- Players join or leave the lobby throughout the lifetime of multiple game sessions.

API Overview

Functions

Operations	Description
CreateLobby	Create and join a new lobby.
FindLobbies	Search for lobbies that fit the provided parameters.
RestoreLobbies	Try to rejoin previously joined lobbies.
JoinLobby	Join a lobby with the provided lobby ID.
LeaveLobby	Leave a joined lobby.
InviteLobbyMember	Invite a player to join a lobby.
DeclineLobbyInvitation	Decline an invitation to join a lobby.
KickLobbyMember	Kick a member from the target lobby.
PromoteLobbyMember	Promote another lobby member to leader. The local player must be the current lobby leader and the target must be the current lobby leader to promote another member.
Modifiers	
MakePublicLobbyData	Change the schema applied to the lobby and member attributes. Only the lobby leader may change the schema. Existing attributes not present in the new schema will be removed.
MakePublicLobbyPolicy	Change the join policy applied to the lobby. Only the lobby leader may change the join policy.
MakePublicLobbyAttributes	Change the attributes applied to the lobby. Only the lobby leader may change the lobby attributes. Attributes are validated against the lobby schema before an update can occur.
MakePublicLobbyMemberAttributes	Change the attributes applied to a lobby member. Lobby members may only change their own attributes. Attributes are validated against the lobby schema before an update occurs.
Accessors	
GetLocalLobbies	Retrieve the list of joined lobbies for the target local player.
GetRecentlyPublishedLobbies	Retrieve the list of recent lobbies for the target local player.
Event Listening	
OnLobbyUpdate	Event triggered when a lobby's data changes.
OnLobbyMemberLeft	Event triggered when a member leaves a lobby.
OnLobbyMemberJoined	Event triggered when a lobby member joins, either as a result of a local player joining or joining a lobby, or when a remote player joins.
OnLobbyMemberAttrChanged	Event triggered when a lobby member leaves a joined lobby.
OnLobbyLeaderChanged	Event triggered when the lobby's leadership changes.
OnLobbySchemaChanged	Event triggered when the lobby's attribute schema changes.
OnLobbyMemberSchemaChanged	Event triggered when a lobby member's attribute schema changes.
OnLobbyInviteReceived	Event triggered when a lobby member receives an invitation.
OnLobbyInviteRemoved	Event triggered when a lobby member withdraws an invitation or the invitation expires.
OnLobbyJoinPolicyChanged	Event triggered when a player requests to join the lobby through an external mechanism.

Enumeration Classes

The Lobbies Interface provides enumeration classes for lobbies. Lobby Join Policy and Lobby Member Leave Reasons through two enumeration classes. ELobbyJoinPolicy and ELobbyMemberLeaveReason.

ELobbyJoinPolicy

Enumerator	Description
PublicAllowInvites	Users can find the lobby through searches based on attribute matching, lobby ID, or invitation.
PublicNoInviteOnly	Users can join the lobby by lobby ID or invitation.
InviteOnlyPolicy	Users can only join the lobby by invitation.

ELobbyMemberLeaveReason

Enumerator	Description
Left	Lobby member chose to leave the lobby.
Kicked	Lobby member was kicked out of the lobby by the lobby owner.
Disconnected	Lobby member left unexpectedly.
Closed	Lobby was destroyed by the online service and all members have left.

Primary Structs

The Lobby Interface functionality is primarily communicated through two structs: FLobbyMember and FLobby. In addition to the function specific structs for passing parameters and return values.

FLobbyMember

Member	Type	Description
AccountId	FAccountId	Account ID of this lobby member.
PlatformUniqueId	FUniqueId	Platform account ID of this lobby member.
PlatformDisplayName	FString	Platform display name of this lobby member.
Attributes	TMap<FSchemaAttributeId, FSchemaAttribute>	Attributes of this lobby member as defined in the configuration.
AllowsPublic	bool	Whether this lobby member is a local player or not. Default value is: false

FLobby

Member	Type	Description
LobbyId	FLobbyId	ID of the lobby.
OwnerAccountId	FAccountId	Account ID of the lobby leader that is the current owner of this lobby.
LocalName	FName	Local name of this lobby.
SchemaId	FSchemaId	ID of the schema applied to this lobby.
MaxMembers	int32	Maximum number of members who can be in this lobby at any given time.
JoinPolicy	ELobbyJoinPolicy	Join policy setting for this lobby.
Attributes	TMap<FSchemaAttributeId, FSchemaAttribute>	Attributes of this lobby as defined in the configuration.
Members	TMap<FAccountId, FMemberMember> FLobbyMembers	Dictionary of lobby members where the key is the member's account ID and the value are probably to that corresponding lobby member struct.

Configuration

Lobbies use a schema system to define the structure and properties of the lobby as well as lobby member attributes. A schema may include multiple lobbies and schema definitions to support many different types of lobbies. All lobby schemas must derive from the LobbyBase schema.

The LobbyBase schema contains all attributes players can use to search for a lobby. This inheritance ensures that all lobbies can be found by other players when using predefined queries to apply to a device search. In Primary Assets Only Lobby is a special attribute within LobbyBase that exists to ensure that if a schema is complete, it can be used by clients.

A game must declare a definition for each schema attribute present in a schema definition. Schemas are defined in a project's configuration files (.ini files). These definitions include an attribute's type, maximum size, and visibility, as well as additional behaviors such as whether the attribute is available in a search parameter.

In general, a schema has categories which contain attribute definitions. The Lobbies Interface schema behavior has categories: Lobby and LobbyMember. These categories contain the attributes which are applicable to the lobby object and lobby member objects, respectively.

Example

For an example configuration for the Lobbies Interface:

DefaultEngine.ini

```
[/Script/OnlineServicesCommon.LobbyConfig_Default]
+Schema=(SchemaId=LobbyBase,bIsAbstract=True,+Attributes=(AttributeId=bPrimaryAssetOnly,Type=Bool,bIsSearchable=True,bIsSensitive=False,bIsVisibleOnClient=True,bIsVisibleOnServer=True,bIsVisibleOnOwner=True,bIsMutable=True,bIsRequiredOnCreate=True,bIsRequiredOnUpdate=True,bIsRequiredOnMigrate=True,bIsRequiredOnJoin=True,bIsRequiredOnLeave=True,bIsRequiredOnKick=True,bIsRequiredOnPromote=True,bIsRequiredOnInvite=True,bIsRequiredOnRemoveInvite=True,bIsRequiredOnDeclineInvite=True,bIsRequiredOnRestore=True,bIsRequiredOnMakePublic=True,bIsRequiredOnMakeLocal=True,bIsRequiredOnMigrate=True,bIsRequiredOnJoinPolicyChange=True,bIsRequiredOnLeaderChange=True,bIsRequiredOnSchemaChange=True,bIsRequiredOnMemberSchemaChange=True,bIsRequiredOnMemberAttributeChange=True,bIsRequiredOnMemberJoin=True,bIsRequiredOnMemberLeave=True,bIsRequiredOnMemberKick=True,bIsRequiredOnMemberPromote=True,bIsRequiredOnMemberInvite=True,bIsRequiredOnMemberRemoveInvite=True,bIsRequiredOnMemberDeclineInvite=True,bIsRequiredOnMemberRestore=True,bIsRequiredOnMemberMakePublic=True,bIsRequiredOnMemberMakeLocal=True,bIsRequiredOnMemberMigrate=True,bIsRequiredOnMemberJoinPolicyChange=True,bIsRequiredOnMemberLeaderChange=True,bIsRequiredOnMemberSchemaChange=True,bIsRequiredOnMemberMemberAttributeChange=True,bIsRequiredOnMemberMemberJoin=True,bIsRequiredOnMemberMemberLeave=True,bIsRequiredOnMemberMemberKick=True,bIsRequiredOnMemberMemberPromote=True,bIsRequiredOnMemberMemberInvite=True,bIsRequiredOnMemberMemberRemoveInvite=True,bIsRequiredOnMemberMemberDeclineInvite=True,bIsRequiredOnMemberMemberRestore=True,bIsRequiredOnMemberMemberMakePublic=True,bIsRequiredOnMemberMemberMakeLocal=True,bIsRequiredOnMemberMemberMigrate=True,bIsRequiredOnMemberMemberJoinPolicyChange=True,bIsRequiredOnMemberMemberLeaderChange=True,bIsRequiredOnMemberMemberSchemaChange=True,bIsRequiredOnMemberMemberMemberAttributeChange=True,bIsRequiredOnMemberMemberMemberJoin=True,bIsRequiredOnMemberMemberMemberLeave=True,bIsRequiredOnMemberMemberMemberKick=True,bIsRequiredOnMemberMemberMemberPromote=True,bIsRequiredOnMemberMemberMemberInvite=True,bIsRequiredOnMemberMemberMemberRemoveInvite=True,bIsRequiredOnMemberMemberMemberDeclineInvite=True,bIsRequiredOnMemberMemberMemberRestore=True,bIsRequiredOnMemberMemberMemberMakePublic=True,bIsRequiredOnMemberMemberMemberMakeLocal=True,bIsRequiredOnMemberMemberMemberMigrate=True,bIsRequiredOnMemberMemberMemberJoinPolicyChange=True,bIsRequiredOnMemberMemberMemberLeaderChange=True,bIsRequiredOnMemberMemberMemberSchemaChange=True,bIsRequiredOnMemberMemberMemberMemberAttributeChange=True,bIsRequiredOnMemberMemberMemberMemberJoin=True,bIsRequiredOnMemberMemberMemberMemberLeave=True,bIsRequiredOnMemberMemberMemberMemberKick=True,bIsRequiredOnMemberMemberMemberMemberPromote=True,bIsRequiredOnMemberMemberMemberMemberInvite=True,bIsRequiredOnMemberMemberMemberMemberRemoveInvite=True,bIsRequiredOnMemberMemberMemberMemberDeclineInvite=True,bIsRequiredOnMemberMemberMemberMemberRestore=True,bIsRequiredOnMemberMemberMemberMemberMakePublic=True,bIsRequiredOnMemberMemberMemberMemberMakeLocal=True,bIsRequiredOnMemberMemberMemberMemberMigrate=True,bIsRequiredOnMemberMemberMemberMemberJoinPolicyChange=True,bIsRequiredOnMemberMemberMemberMemberLeaderChange=True,bIsRequiredOnMemberMemberMemberMemberSchemaChange=True,bIsRequiredOnMemberMemberMemberMemberMemberAttributeChange=True,bIsRequiredOnMemberMemberMemberMemberMemberJoin=True,bIsRequiredOnMemberMemberMemberMemberMemberLeave=True,bIsRequiredOnMemberMemberMemberMemberMemberKick=True,bIsRequiredOnMemberMemberMemberMemberMemberPromote=True,bIsRequiredOnMemberMemberMemberMemberMemberInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberRemoveInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberDeclineInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberRestore=True,bIsRequiredOnMemberMemberMemberMemberMemberMakePublic=True,bIsRequiredOnMemberMemberMemberMemberMemberMakeLocal=True,bIsRequiredOnMemberMemberMemberMemberMemberMigrate=True,bIsRequiredOnMemberMemberMemberMemberMemberJoinPolicyChange=True,bIsRequiredOnMemberMemberMemberMemberMemberLeaderChange=True,bIsRequiredOnMemberMemberMemberMemberMemberSchemaChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberAttributeChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberJoin=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberLeave=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberKick=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberPromote=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberRemoveInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberDeclineInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberRestore=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMakePublic=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMakeLocal=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMigrate=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberJoinPolicyChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberLeaderChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberSchemaChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberAttributeChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberJoin=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberLeave=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberKick=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberPromote=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberRemoveInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberDeclineInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberRestore=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMakePublic=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMakeLocal=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMigrate=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberJoinPolicyChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberLeaderChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberSchemaChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberAttributeChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberJoin=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberLeave=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberKick=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberPromote=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberRemoveInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberDeclineInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberRestore=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMakePublic=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMakeLocal=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMigrate=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberJoinPolicyChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberLeaderChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberSchemaChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberAttributeChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberJoin=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberLeave=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberKick=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberPromote=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberRemoveInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberDeclineInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberRestore=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMakePublic=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMakeLocal=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMigrate=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberJoinPolicyChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberLeaderChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberSchemaChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberAttributeChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberJoin=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberLeave=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberKick=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberPromote=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberRemoveInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberDeclineInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberRestore=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMakePublic=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMakeLocal=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMigrate=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberJoinPolicyChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberLeaderChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberSchemaChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberAttributeChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberJoin=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberLeave=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberKick=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberPromote=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberRemoveInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberDeclineInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberRestore=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMakePublic=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMakeLocal=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMigrate=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberJoinPolicyChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberLeaderChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberSchemaChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberAttributeChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberJoin=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberLeave=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberKick=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberPromote=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberRemoveInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberDeclineInvite=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberRestore=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMakePublic=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMakeLocal=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMigrate=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberJoinPolicyChange=True,bIsRequiredOnMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberMemberLeaderChange=True,bIsRequiredOnMember
```